

The paper builds on the Assembly Calculus (AC) model, introducing more realistic elements, making it more biologically plausible. In the original AC model, the k-winners-take-all (k-WTA) algorithm was used as a form of inhibition to control the size of formed neuronal assemblies. Here, the authors first replace the k-WTA algorithm with the more biologically grounded E%-winner-take-all, inspired by gamma frequency oscillation dynamics, and additionally introduce inhibitory synapses. The motivation behind this work is reasonable, and the results suggest potential improvement in certain qualities of the formed assemblies (e.g. size and density) and model performance, especially in memory recall (assembly reactivation) compared to the original AC model, with potential importance to researchers in the field. However, in its current state, the manuscript has significant presentation issues, the biological insights are barely addressed, and several claims are questionable given the experiments provided. Clearer presentation of the methodology and further experiments are required to properly evaluate the validity of the claims.

Major concerns:

The introduction is superfluous, and the language is not very concise. A lot of the information presented here is irrelevant to the work being done, while relevant information is missing. The same can be said for the treatment of the model and its features. Some topics that should be introduced here are, for example, “biological inhibition”, which is the novel element of this work. Also, the motivation and rationale behind why more than one type of inhibition is introduced (or required) is also absent.

The methods are not entirely clear. There are a lot of interconnected pieces, and a lot of back and forth is required in order to understand everything. Better presentation is therefore necessary. A lot of information is placed when it is not needed and repeated later, complicating comprehension. There is no clear distinction between what belongs to the original model and what is new, and much of this had to be derived implicitly. It would be better if the proposed model is presented first and, at the end of each sub-section, the differences with AC, where they exist. If the differences are only in parameter values, those should be given in Table 1 only. Most importantly, some information is missing. For example, details on how structural plasticity is implemented, which synapses are plastic and which exhibit structural plasticity, which process decides which newly formed synapses become excitatory or inhibitory, and what the recovery protocol is.

For most of the experiments in the results section, a clear motivation and rationale are missing. The authors evaluate the model on criteria like the number of iterations required for assembly formation, the size and synaptic density of the formed assemblies, and compare these to the original AC model. However, it is unclear what this comparison aimed to achieve, and some of the authors' claims are not properly or clearly backed up by the results. More generally, many of the results show differences between the proposed model and the original AC model, but these differences do not always reveal whether and why the proposed model is better.

Importantly, the authors use the E%-WTA algorithm as an alternative to k-WTA, but this introduces a novel problem (large assembly sizes) not present in the original AC model. To

compensate for this, “feedforward inhibition” is added as an additional inhibitory mechanism. Yet, the contribution of this mechanism seems already quite significant by itself, since it can reduce the number of active neurons considerably. In fact, it seems to be one of the most impactful changes they introduced (see Figure 3a + Table 2). However, the authors do not evaluate whether feedforward inhibition alone can achieve similar control without the E%-WTA algorithm. Specifically, I cannot see why the conditions the authors established for the formation of assemblies should not work in the absence of the E%-WTA algorithm, using feedforward inhibition alone.

As a result of the above, the impact of the current version is questionable. It would be useful to know the maximum capacity of formed assemblies in the proposed model and how that number compares to the original AC model. How does the number of formed assemblies affect the recall process relative to the AC? These are partly addressed in “Limitations” (Figure 4), however, there is no comparison with the original AC model, making it hard to evaluate whether the introduced changes are beneficial. Incorporating this analysis into the main results would strengthen the contribution of the work.

Lastly, the discussion does not fully address the implications of the results. For example, the differences between the two models (in assembly size, synaptic density, etc) are not discussed in relation to biology, and their implications are left largely unaddressed.

Specific comments:

- (Lines 41-44) This sentence introduces sequential activation of assemblies as a means of information propagation. However, the paper doesn’t address sequential propagation at all.
- (Lines 56-60) This is a very broad sentence. “considerable advances in the field of neuroscience” is very vague. “our comprehensive understanding of neuronal function at the cellular and molecular level” is an overly optimistic statement (that is also not supported by any references here).
- The description of the AC model (lines 61-68) here is too detailed and should be moved into the results or methods (wherever the authors decide to describe the two models) in detail. The important content here is what the AC was proposed to address, and that it uses assemblies formed through Hebbian learning as the functional unit.
- (Lines 68-70) This is useful information. However, the last sentence is left without a brief explanation of what these “functional properties are”, so the relevance to this work is not clear.
- (Lines 83-87) The E%-WTA algorithm is mentioned as a more realistic mechanism for inhibition. However, the motivation and background are barely explained. It might be helpful to support the selection of E%-WTA by explaining why it would address the problems of k-WTA (previous paragraph).
- (Lines 89-92) The rationale for selecting these properties to evaluate the model is unspecified. Also, the practical meaning and impact of these properties and why they matter are never stated.

- (Lines 97-98) Half of this sentence talks about synapse formation, and the other half about neuronal projections. Please keep sentences compact.
- (Lines 116-123) This information should be moved to the introduction to motivate the E%-WTA algorithm.
- Equation 4. If I understand correctly, the value of E is a time-dependent value and is specific for each neuron. Then $E_j(t)$ would be more appropriate.
- (Line 132) What is the algorithm's sensitivity to the parameters τ_m and d ?
- (Lines 143-148) Again, much of this information should be moved to the introduction for motivation. The E-I balance can be explained at multiple levels. "Balance between excitatory and inhibitory stimuli" is an ill-defined term.
- (Lines 160-175) The authors use lines 160-162 to justify the introduction of three conditions. However, the first two conditions they introduce already solve the problem of lines 160-162. So the incorporation of the third condition seems unjustified. Moreover, the current formulation addresses only the difference of one consecutive timepoint (t and $t-1$). In Figure 2b, however, they compare this to multiple timepoints in the future and for each consecutive timepoint the number and identity of neurons keep changing. How do these three conditions account for this? What is the motivation for the "higher synaptic density" condition?
- (Lines 203-209) This information is unnecessary here since it is presented in detail in the results.
- Figure 1f, and particularly the right plot, is not informative. I cannot understand what information it conveys.
- Please keep parameter values summarised in Table 1 and avoid using them in the text unless needed (e.g. lines 193-194).
- In Table 1, w_{inh} has an asterisk that is not addressed anywhere.
- (Lines 225-226) If the density condition is by definition a requirement for an assembly to form, then how can there be formed assemblies that do not fulfil this requirement? Please clarify.
- (Lines 232-233) Why was the number of neurons for the minimum size of an assembly set to 6? This seems like an arbitrary choice. What was the rationale behind it?
- (Lines 233-237) How many simulations were run in total? For which parameter values specifically? Figure 2 shows 5 values for w_{inh} and 5 for β . Then in the caption it says 100 for each parameter value. Does that mean that $5 \times 5 \times 100 = 2500$ simulations in total? Why do some of these lead to large failures? What was the reason for the failure? Was it due to random initialisation and random selection of the input neurons?
- (Lines 239-283) It is not clear why synaptic density is a good measure for comparison. Which density is a "good density" and why? Is higher density always better? Why? Please clarify.
- (Line 241) Here, the parameter β is referred to as "synaptic plasticity". However, in the methods it is introduced as "learning rate", which is more accurate. Please be concise.
- (Lines 249-251) This seems like a negative result since k-WTA achieves formation with fewer iterations. Although this is not exactly a fair comparison since the conditions are

different, it still seems negative. What are the implications of this, especially considering that previously you showed that a lower beta is better for assembly formation?

- (Lines 251-253) The formation was slightly slower, but the E%-WTA requires more iterations. I cannot see how this claim here can hold based on these contradictory results. Please clarify.
- (Line 254) The E%-WTA abbreviation was already introduced before. Please just use the abbreviation.
- (Line 260) The symbol "A" was already introduced as a "brain area" in the methods and re-used here for "neural assembly". This creates ambiguity.
- (Line 285-298) Arguably the strongest result. I would emphasise this more in the introduction and abstract. Although how this is measured should be clearer. How is recovery defined? Does it require the "formed assembly" to be activated upon stimulus presentation? Does the assembly need to be recovered when activated, only by itself? What if other neurons are effectively activated in addition to those that belong to the assembly? Please provide more information.
- (Lines 299-309) This seems like rather an unfair comparison. The kWTA has fixed-size assemblies, and the feedforward inhibition was already optimised to keep the number of neurons in assemblies in E%-WTA lower than that. Shouldn't this by default produce smaller overlap? For the discussion: What are the implications of this reduced overlap?
- (Lines 310-319 and Figure 4) In this section, some useful insights are given as to why the model fails under certain conditions. The manuscript would become considerably stronger if the authors expanded more on these and made them part of the results.
- Line 311. The authors mention that the proposed model "outperforms" the existing one. However, such a claim is not evident from the previous results. Differences between the two models exist, but why these differences seem to show that the proposed model is better is unclear.
- Figure 3b. The legend has blue colour for the E%-WTA model. However, only the outliers are visible. Please consider using coloured outliers in the plots.
- Table 2. This table, although the most informative piece in the paper, is quite hard to understand fully. Please consider presenting the results in the figures (at least the most significant ones) and leaving the details/summary for the Table. It is unclear to me what exactly each superscript letter (a, b, c...) refers to specifically. Please clarify.
- "neuronal group" and "neuronal assembly" have been used interchangeably throughout the text, creating confusion. Please use only one term.
- Please check thoroughly for grammatical mistakes (e.g. "sucess" in Figure 4, "dynamic" in Line 101, "your" in Line 105)
- The titles of each section are not really informative. They should summarise the main result of each section. Also, each section on the results should start with some rationale for what this section addresses.
- Please check the references thoroughly and make sure they are correct. For example, reference [25] reads: "Anonymous. Statistical mechanics for networks of real neurons. Rev Mod Phys. 2025 Apr."
- In all figures, the labels, legends, tick labels, and more or less all text are too small to read.